

CHE 313

Quiz 3

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$$F = 0.6 \text{ kmol} \quad X_F = 0.7 \quad X_{W, \text{final}} = 0.1$$



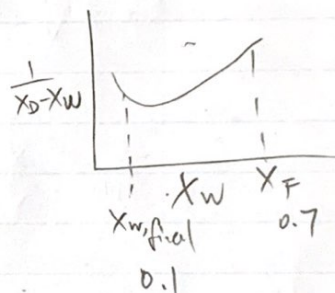
1) M.B. : $F = W + D$

Component M.B. : $X_F \cdot F = X_W \cdot W + D \cdot X_D$

2) Find W_{final} , D_{tot} , $X_{D, \text{avg}}$

$$\ln \frac{W_{\text{final}}}{F} = - \int_{X_{W, \text{final}}}^{X_F} \frac{dx_W}{X_D - X_W}$$

$$\frac{W_{\text{final}}}{F} = e^{-\text{Area of } \left(\frac{1}{X_D - X_W} \text{ vs } X_W \right) \Big|_{X_{W, \text{final}}}^{X_F}}$$



$$\text{Area} = \frac{(0.7 - 0.1)}{6} \left[\frac{1}{0.418 - 0.1} + \frac{4}{0.729 - 0.4} + \frac{1}{0.87 - 0.7} \right]$$

$$= 0.1 (3.144654088 + 12.15805471 + 5.882353)$$

$$= 0.1 (21.18506174) = 2.1185$$

$$\frac{W_{\text{final}}}{F} = e^{-2.1185}$$

$$W_{\text{final}} = F \cdot e^{-2.1185} = 0.6 \cdot 0.1202110689 = 0.0721266413 \text{ kmol}$$

$$D_{\text{tot}} = F - W_{\text{final}} = 0.5278733587 \text{ kmol} = D_{\text{tot}}$$

$$X_{D, \text{avg}} = \frac{X_F \cdot F - X_W \cdot W}{D} = \frac{0.7 \times 0.6 - 0.1 \times 0.0721266413}{0.5278733587} = 0.7819817558$$